

Testing Plan Document

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November 24, 2020

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Purpose

The purpose of this application is to teach young children the alphabet and build their vocabulary via interactive games.

Features to be Tested

Base States

i)

Name	Learning Base
Requirement	N/A
Preconditions	The system is turned on.
Steps	1. Power on the system 2. Press any alphabet button
Expected Results	Pressing any alphabet button results in a corresponding letter and sound to play.

ii)

Name	Quiz Base
Requirement	N/A
Preconditions	The system is turned on.
Steps	1. Power on the system 2. Press the switch mode button
Expected Results	The quiz will start and the first question will be asked.

Software Test Cases

i)

Name	Verify that the user can switch between “Learning Mode” and “Quiz Mode”
Requirement	R001. The system will have two different modes: “Learning” and “Quiz”
Preconditions	The system is turned on and in Learning Base
Steps	1. Press the “Quiz Mode” button in Learning Mode. 2. Press the “Learning Mode” button in Quiz Mode.
Expected Results	The system should switch from Learning Mode to Quiz Mode, and vice versa.

ii)

Name	Verify that the learning mode splash screen displays properly
Requirement	R002 Learning mode must display a splash screen when it starts
Preconditions	System is turned on or quiz mode has just finished
Steps	1. User presses the power button
Expected Results	The splash screen will display for 3 seconds before allowing the user to input letters

iii)

Name	Verify that each alphabet button displays an image and sound
Requirement	R003 Learning Mode will allow the user to press each alphabet button and listen to how the letter sounds.
Preconditions	The system is turned on and in Learning Base
Steps	1. Press the first alphabet button of A 2. See that the system displays the proper image and plays the proper sound 3. Repeat the previous step for the next alphabet buttons
Expected Results	Each alphabet button should display the corresponding image and play the corresponding sound.

iv)

Name	Splash screen for quiz mode displays
Requirement	R004 Quiz Mode will display a splash screen before starting with the first question
Preconditions	System is on and user has just pressed the quiz mode button
Steps	1. User presses the quiz mode button 2. The splash screen displays for 3 seconds
Expected Results	The system will wait on the splash screen for 3 seconds and then the first question will display

v)

Name	Verify that users can answer questions properly
Requirement	R005 Software must accept an input from the user (quiz mode)
Preconditions	The system is in quiz mode and a quiz has been started.
Steps	1. User presses an alphabet button.
Expected Results	The system displays a message corresponding to whether the input was correct or not, and continues with the quiz.

vi)

Name	Verify that the user can see their score on the quiz
Requirement	R006 Software must return a final score for a quiz in Quiz Mode
Preconditions	System is on the final question of a quiz
Steps	1. User enters input for final question.
Expected Results	System accepts input and displays the final score.

Hardware Test Cases

i)

Name	Verify that rotating the case does not impede operation of the toy
Requirement	HR001 Case and buttons must be able to rotate around base
Preconditions	All hardware including the case, buttons, speakers, Arduino and Raspberry Pi must be assembled
Steps	1. Rotate the top half of the case fully around to the left stopper 2. Rotate the top half of the case fully around to the right stopper
Expected Results	All hardware will still be operational no matter where the case is rotated to

ii)

Name	Verify the power button works properly
Requirement	HR002 System must be able to turn off/on
Preconditions	The power button must be assembled and plugged in
Steps	1. Power on the machine 2. Power off the machine
Expected Results	The power button will turn off and on all components of the toy

iii)

Name	Verify that all alphabet buttons are assembled and plugged in properly
Requirement	HR003 System must contain 26 alphabet buttons
Preconditions	All alphabet buttons are assembled and plugged in
Steps	1. Check the wiring on all alphabet buttons to make sure nothing has come unplugged
Expected Results	All alphabet buttons will work properly

iv)

Name	Verify that holographic images are visible and readable
Requirement	HR004 System must display holographic image
Preconditions	The holographic images are loaded on the Raspberry Pi, and the code can access them
Steps	1. Test all operations that cause a holographic image to display
Expected Results	The holographic images will be large enough to read.

v)

Name	Verify that audio is clear and audible
Requirement	HR005 System must be able to output audio
Preconditions	The speakers are plugged in and the code to output audio works properly.
Steps	1. Perform all actions that would cause audio to output
Expected Results	Audio will be clear and at an acceptable volume.

Features Not to be Tested

- Software must correlate an image to a word (quiz mode)- This functionality is hardcoded into the software.
- Software must correlate audio to a word (quiz mode)- This functionality is hardcoded into the software.
- Software must draw random entries from a word bank (quiz mode)- The design of quiz mode has been changed to consist of a fixed set of questions pertaining to a certain topic.
- Software must be able to implement addition of new and different word banks (quiz mode)- This feature has been removed due to a lack of time to implement it.

Approach

- Each test case will be run 3 times to ensure the correct result was achieved.
- Each test case will be tested on its own to ensure no outside influence tampers with the results of the test.
- Each test case will be performed again when new code or hardware is added to ensure that what was previously working continues to function properly.
- All team members will strive to resolve any issues discovered by test cases, whether by fixing the issue or changing the project plan to work better.

Environmental Needs

- Arduino Mega 2560
- Raspberry Pi 4B running Raspberry Pi OS
- 10" Display screen
- Plexiglass that is the holographic environment
- 26 Alphabet buttons each on mini breadboards
- 2 Mode buttons
- 1 Power button
- USB power bank

Schedule

- **Nov 21:**
 - Unit testing for letter buttons
- **Nov 25:**
 - Quiz Mode code completed
 - Insert buttons into body of model
 - Test buttons in model
 - Test Quiz Mode
 - Test entire product
- **Nov 28:**
 - Extra day for testing (if needed)
- **Dec 1:**
 - Present final product

Acceptance Criteria

All test cases in this document must be met. All team members must agree that the product is ready to turn in. The hardware must be assembled in the 3D printed case and tested for functionality after assembly. Source code must be documented, commented, and organized into methods if not already written in such a manner (Main method should not contain excess code).

Roles and Responsibilities

The software team, consisting of Khoi, Sean, and Syed will be responsible for finishing the code, completing all the test cases for the project's software requirements, and revising the code based on the outcomes.

The hardware team, consisting of Andy and John will be responsible for completing the hardware components and shell, completing all the test cases for the project's hardware requirements, and improving the hardware based on the outcomes.